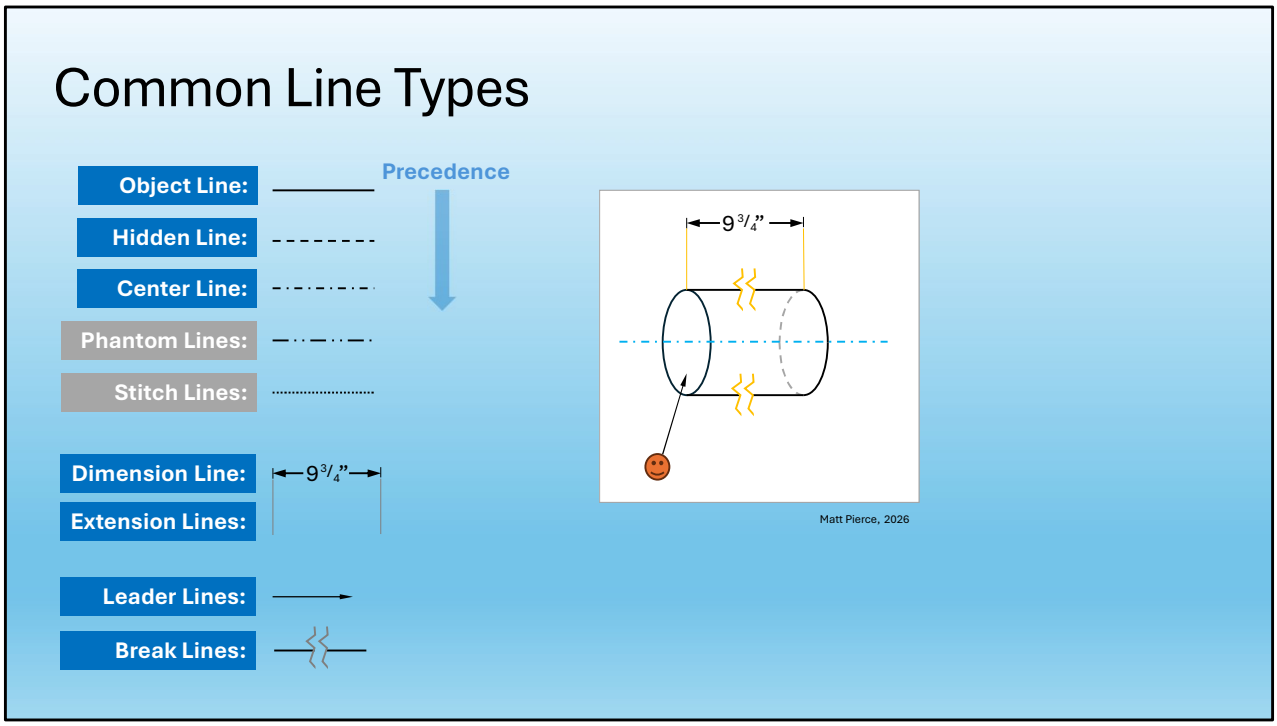
Aerospace Manufacturing Core
Study Guide

"AMCSG-303"

Blueprints & Documents

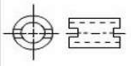
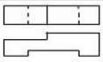
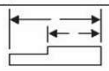
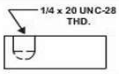
Lines & Symbols

Line Styles



- ChatGPT: make a sample revision block from blueprint for aerospace manufacturing

USNavy14040A - Standard Line Types (1 of 2)

LINE STANDARDS			
Name	Convention	Description and Application	Example
Center Lines		Thin lines made up of long and short dashes alternately spaced and consistent in length. Used to indicate symmetry about an axis and location of centers.	
Visible Lines		Heavy unbroken lines Used to indicate visible edges of an object	
Hidden Lines		Medium lines with short evenly spaced dashes Used to indicate concealed edges	
Extension Lines		Thin unbroken lines Used to indicate extent of dimensions	
Dimension Lines		Thin lines terminated with arrow heads at each end Used to indicate distance measured	
Leader		Thin line terminated with arrowhead or dot at one end Used to indicate a part, dimension or other reference	

(PD) U.S. Gov – NAVEDTRA14040A, 2015

- ChatGPT query identified USGovernment public domain document: https://media.defense.gov/2017/Nov/03/2002658619/-1/-1/1/171103-N-ZY182-15347.pdf
  Figure 2-21 on page 33

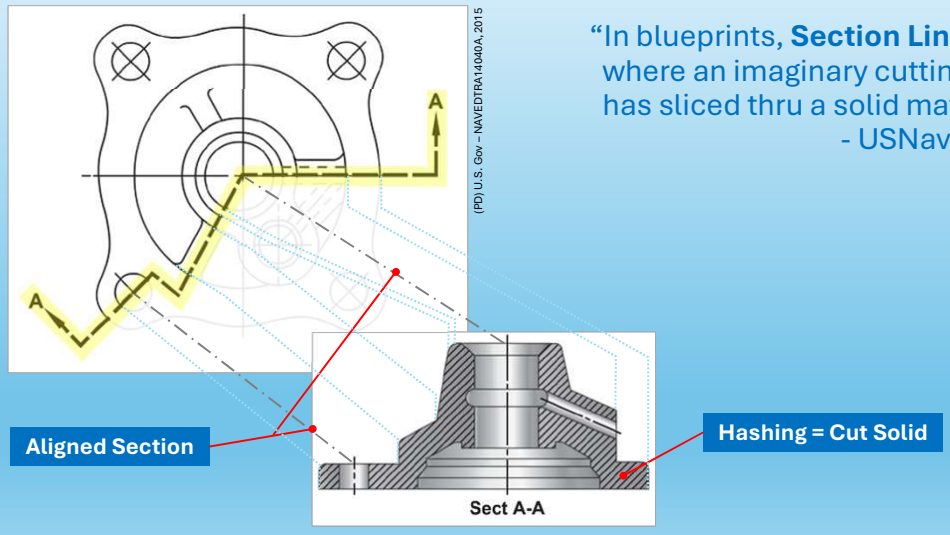
USNavy14040A - Standard Line Types (2 of 2)

Name	Convention	Description and Application	Example
Break (Long)		Thin, solid ruled lines with freehand zigzags Used to reduce size of drawing required to delineate object and reduce detail	
Break (Short)		Thick, solid free hand lines Used to indicate a short break	
Phantom or Datum Line		Medium series of one long dash and two short dases evenly spaced ending with long dash Used to indicate alternate position of parts, repeated detail or to indicate a datum plane	
Stitch Line		Medium line of short dases evenly spaced and labeled Used to indicate stitching or sewing	Stitch
Cutting or Viewing Plane Viewing Plane Optional		Thick solid lines with arrowhead to indicate direction in which section or plane is viewed or taken	
Cutting Plane for Complex or Offset Views		Thick short dashes Used to show offset with arrowheads to show direction viewed	

(PD) U.S. Gov – NAVEDTRA14040A, 2015

- ChatGPT query identified USGovernment public domain document: https://media.defense.gov/2017/Nov/03/2002658619/-1/-1/1/171103-N-ZY182-15347.pdf
  Figure 2-21 on page 33

Section Lines



(PD) U.S. Gov – NAVEDTRA14040A, 2015

(PD) U.S. Gov – NAVEDTRA14040A, 2015

"In blueprints, **Section Lines** indicate where an imaginary cutting plane has sliced thru a solid material."
- USNavy 14040A

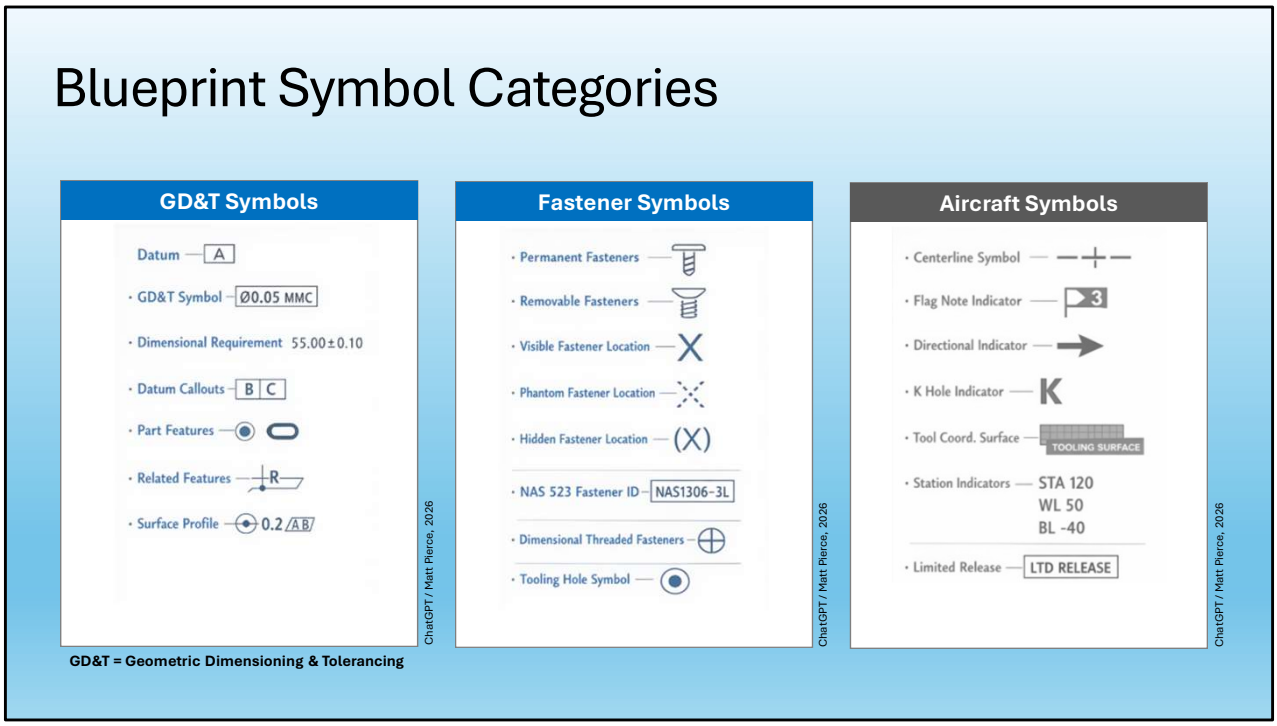
- ChatGPT query identified USGovernment public domain document: https://media.defense.gov/2017/Nov/03/2002658619/-1/-1/1/171103-N-ZY182-15347.pdf
  – page 39, Figure 2-35

Blueprint Symbols

Blueprint Symbols

"Blueprint symbols are standardized universal marks
used as shorthand code
to save space, reduce errors, and
provide clear, concise, and consistent information
across technical drawings."

- Dr. Google

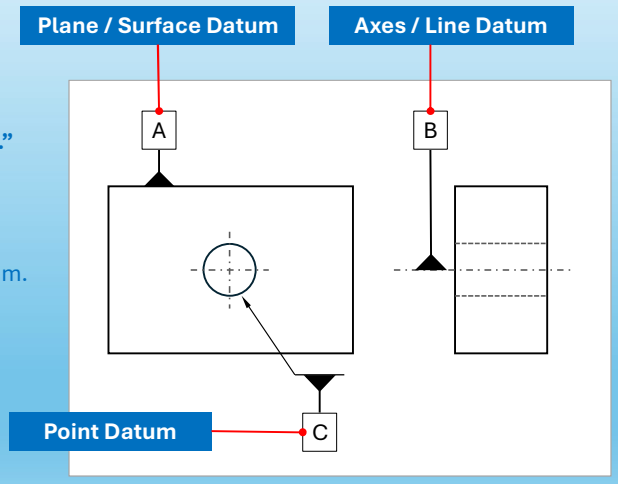


- ChatGPT:
  "make a diagram comparing three different aerospace manufacturing blueprint symbol categories:
  1. geometric symbols: datum, GD&T symbol, dimensional requirement, datum callouts, part features, related features, surface profile
  2. fastener symbols: permanent fasteners, removable fasteners, visible fastener location, phantom fastener location, hidden fastener location, fastener identification using NAS 523 standard and four quadrants, dimensional threaded fasteners, and tooling hole symbol
  3. aircraft symbols: centerline symbol, flag note indicator, directional indicator, K or coordinating hole indicator, engineered tooling coordinated surface, station indicators (STA,WL, BL) and numbers identifying distinct from 0, and limited release indicator"

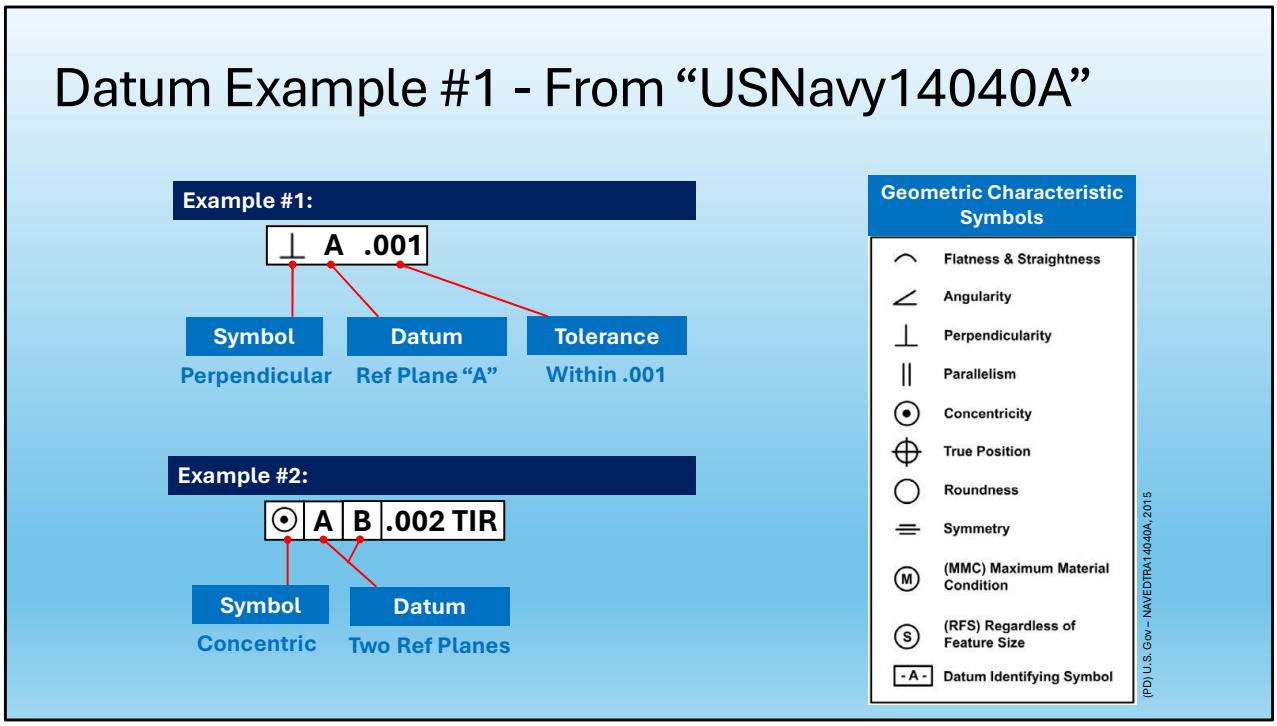
Datum Definition

"A datum is...
a surface, line, or point
from which a geometric position
is to be determined
or from which a distance is to be measured."
- US Navy "NAVEDTRA 1404A"

Start measuring or dimensioning from the datum.

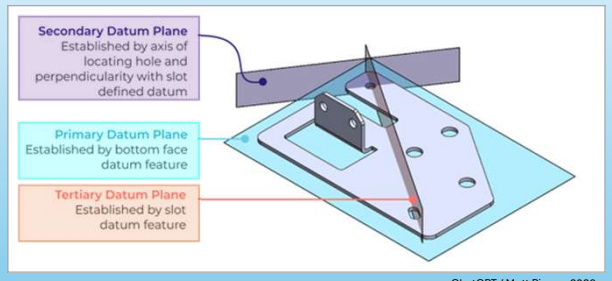


Matt Pierce, 2026

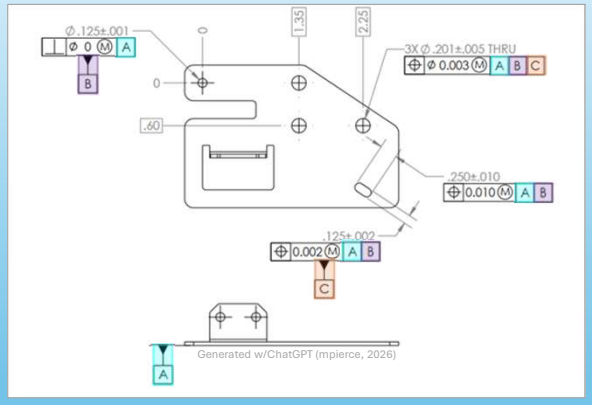


- ChatGPT query identified USGovernment public domain document: https://media.defense.gov/2017/Nov/03/2002658619/-1/-1/1/171103-N-ZY182-15347.pdf
(pg 75)

Datum Example #2 - From “ChatGPT”

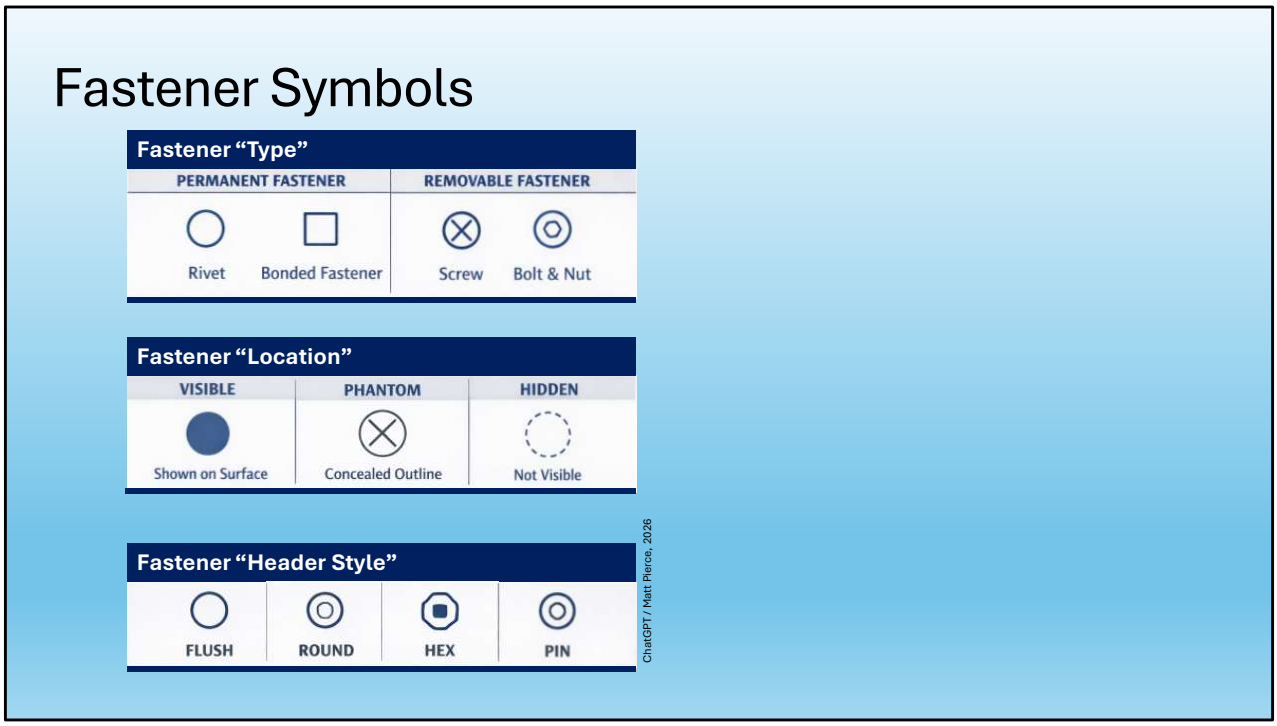


ChatGPT / Matt Pierce, 2026

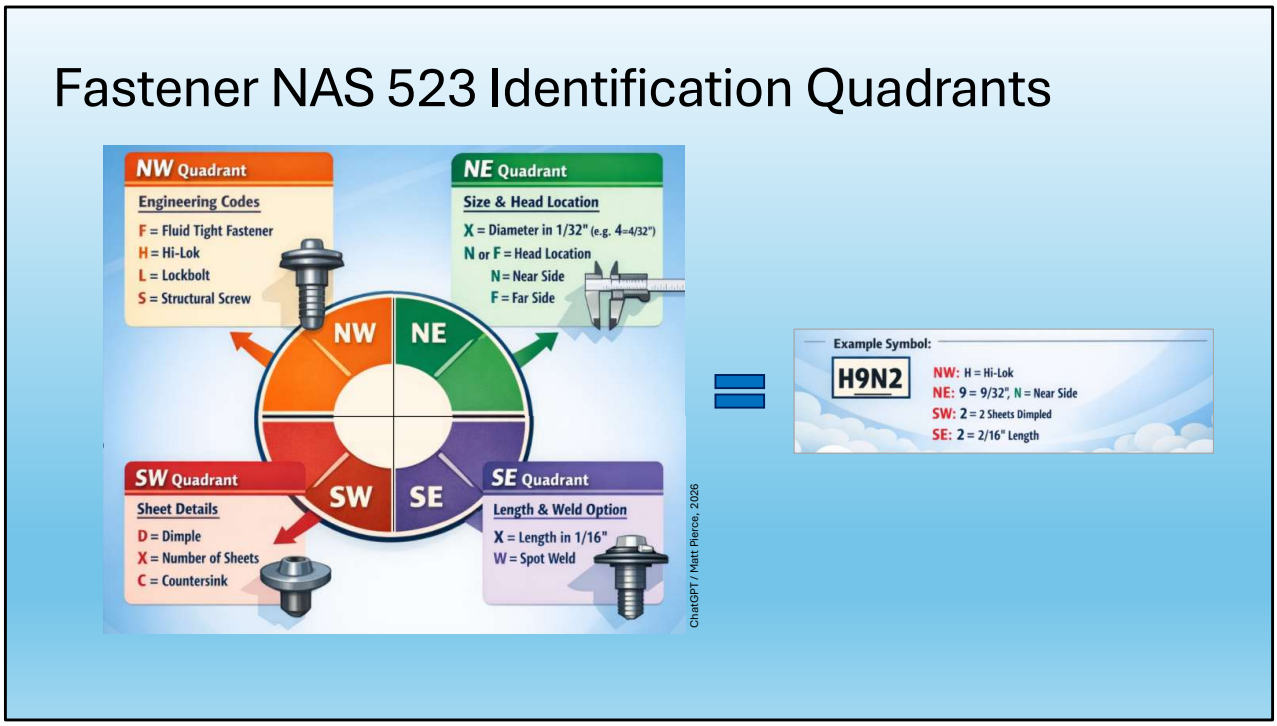


ChatGPT / Matt Pierce, 2026

- ChatGPT: make a diagram showing what a datum, datum feature, datum feature letters, and datum line are.



- ChatGPT: make an infographic showing aerospace fastener symbols: permanent vs. removable fastener, visible vs. phantom vs. hidden fastener location, and anything else appropriate for the chart
  + make them all 2 dimensional, top-down view drawing symbols (same as how "visible", "phantom", and "hidden" fastener locations are done.)



ChatGPT: make an infographic about "National Aerospace Code 523", specifically it's method of symbolically identifying fasteners thru four quadrants "NW, NE, SW, and SE".
1. NW quadrant shows engineering codes for the fastener (eg: fluid tight fastener, etc.)
2. NE quadrant shows two identification codes (X = diameter of fastener in 1/32s of inch, and N or F as the location of the manufactured head)
3. SW quadrant shows three identification codes (D = Dimple, X = number of sheets to be dimpled, and C = Countersink)
4. SE quadrant shows two identification codes (X = fastener length in 1/16s of an inch, and W = spot weld option)

- ChatGPT: make an infographic about NAS 523 symbolic instructions for identifying fasteners. Specifically identify the NW, NE, SW, and SE quadrants. Make it a training poster format.
  + remove the center circle "NAS 523 Symbol". Make the four white dividing lines intersect in the middle.
  + Remove the arrowheads from the horizontal gray line in middle of chart.

Remove title "Symbolic Instructions for Identifying Fasteners“
+ Remove label "SW QUADRANT". Remove label "SE QUADRANT. Remove label "NE QUADRANT"

Q&A

Practice Q&A's (Q1-Q8)

Q1: Name of solid lines that depicts the shape of the part or assembly?

A: Object Lines

Q2: Name of dashed lines that depict hidden surfaces, edges, or hole walls?

A: Hidden Lines

Q3: Name of dash-dot-dash lines that indicate the middle, or symmetric on both sides?

A: Center Lines

Q4: Name of lines that have numeric lengths shown (in inches typically)?

A: Dimension Lines

Q5: Thin lines that drop away 90° from the Dimension Line arrow points to the points they are measuring?

A: Extension Lines

Q6: Cut lines showing on one view what a different view is seeing with a section removed?

A: Section Lines

Q7: Name of standardized universal marks used to save space, reduce errors, and provide consistent info on technical drawings?

A: Blueprint Symbols

Q8: Name of blueprint symbols encompassing datums, part features, surface profiles, etc.?

A: GD&T Symbols: Geometric Dimensioning & Tolerancing

Practice Q&A's (Q9-Q16)

Q9: Name of blueprint symbols encompassing fastener details?

A: Fastener Symbols

Q10: Name of blueprint symbols encompassing station indicators, tool surface, flag notes, etc.?

A: Aircraft Symbols

Q11: Name of a surface, a line, or a point that is a starting reference position?

A: Datum

Q12: What quadrant of the NAS 523 shows the engineering fastener code (eg: fluid tight fastener)?

A: NW

Q13: What quadrant of the NAS 523 shows the fastener diameter and location (near/far)?

A: NE

Q14: What size does the NE quadrant fastener diameter "7" represent?

A: 7/32". Number 7 is count of 1/32"s. Diameter = shorter = smaller 1/32.

Q15: What quadrant of the NAS 523 shows dimple, countersink, and # of sheets?

A: SW

Q16: What quadrant of the NAS 523 shows length and spot weld?

A: SE

Practice Q&A's (Q17)

Q17: What size does the SE quadrant fastener length "8" represent?

A: 8/16". Number 8 is count of 1/16". Length = longer = bigger 1/16.

Thank You

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https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg